

OSI Model

The Open Systems Interconnection (OSI) model, also called the OSI reference model is a conceptual model that divides network communication and interoperability into seven abstract layers.



These layers are:

Layer 7: The application layer

It initiates communication with the network, including the protocols and data manipulation processes that convert computer-readable network data into user-readable responses. It's responsible for processes such as:

- File transfer
- Communication and authentication
- Remote access
- Directory services

Layer 6: The presentation layer

It prepares data for the application layer, including data translation, compression and encryption. It supports secure sockets layer/transport layer security (SSL/TLS) protocols, JPEG protocols (for image compression) and MPEG protocols (for video compression)

Layer 5: The session layer

It initiates and terminates connections between two devices interacting on the network, making sure that resources are neither overused nor underutilized. Key functions are:

- Session interactions
- Synchronization
- session recovery

Layer 4: The transport layer

it transmits end-to-end data between two devices interacting on the network, making sure that data isn't lost, misconfigured or corrupted. The transport layer also handles:

- Service point addressing
- flow control
- multiplexing

Layer 3: The network layer

it handles data addressing, routing and forwarding processes for devices interacting across different networks. If the devices are on the same network, they don't need the network layer to interact. The network layer primarily uses the Internet Protocol v4 (IPv4) and IPv6 and is responsible for:

- Packet fragmentation and reassembly,
- Traffic control

Layer 2: The data link layer

it manages data routing between two interacting devices on the same network. When the DLL receives a packet from the network layer, it divides the packet into data "frames" according to the frame size of the network interface card (NIC) and transmits it to the host using its MAC address. It also includes:

- Framing
- physical addressing
- error control
- flow control
- access control

Layer 1: The physical layer

it comprises the physical assets, like routers and USB cables, that convert data into strings of 1s and 0s for transmission to higher layers. It is also responsible for:

- bit rate control
- Bit synchronization
- Transmission mode
- Physical topologies.

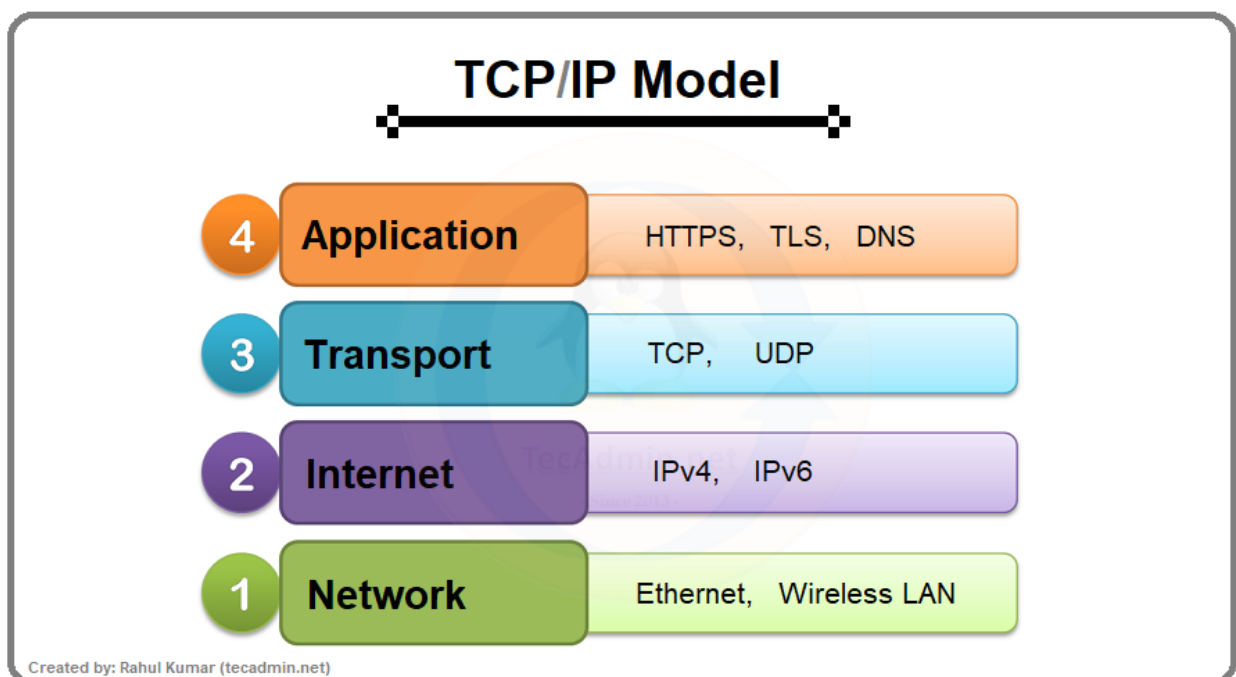
TCP/IP Model

Tcp: The Transmission Control Protocol is a communications standard that relies on an established connection between client and server before information, or data, is transmitted.

IP: The Internet Protocol is responsible for finding the IP address, which is the digital location of the device where the data is being transmitted. Any device on a particular network is identified by the series of numbers assigned to it known as an IP address.

TCP/IP:

The main responsibility of TCP/IP is the transferring and receiving of data of a computer between two devices. It makes sure the data is reliable and error-free. It is possible to separate TCP/IP, but the functions are used together so often that they make more sense coupled together.



Layers of the TCP/IP Model

There are four defined layers of the TCP/IP model:

- 1. Application Layer:** The application layer is responsible for maintaining a smooth connection for data exchange between the user and the application. It is the final stage of the TCP/IP model and allows information to be exchanged once the connection has been established by the other layers of the protocol. Its protocols include HTTP, FTP, Post Office Protocol 3 (POP3), SMTP, DNS, Dynamic Host Configuration Protocol and SNMP.
- 2. Transport Layer:** The transport layer is the part of the TCP protocol responsible for setting up the basic data channels for task-specific communication. It implements host-to-host connectivity for reliable message transfer services. It provides:
 - Error control: Allowing data to be transferred without problems
 - Segmentation: The breaking up and organizing of data packets
 - Flow control: Maintains the speed in which data is sent
 - Congestion control: Seeks out and distributes data on the most efficient path to the destination
 - Application addressing (or port numbers): This provides devices with a destination address for the data packets
- 3. Internet Layer:** The internet layer, also called the network layer, is part of the IP protocol. It controls where the data packets are sent in the connected network. However, the internet layer does not recognize the different transport layer protocols, and as a result, it can carry data for a variety of different upper layer protocols.
- 4. Network Access Layer:** is defined by the physical parts of the computer or device that send and receive data. These include but are not limited to:
 - Ethernet cables
 - Wireless networks
 - Network interface cards
 - Device drivers in the computers

COMMON NETWORK PORTS

Common ports	Ports numbers	Used for
HTTP	80	Used for web communication
HTTPS	443	Secure version of HTTP
FTP	20 and 21	Transfers files between systems

DHCP	67 and 68	Automatically assigns IP addresses
DNS	53	Converts domain names into IP addresses
SNMP	161/162	Monitor and manage devices
ICMP	NO PORT	Network diagnostic
NTP	123	Time synchronization
POP3	110	Email retrieval
IMAP	143	Email access
TELNET	23	Unencrypted remote access

What is an IP Address?

An IP address uniquely identifies a device on a network. IPv4 consists of 32 bits divided into four octets. Example: 192.168.x.x

Public IP Address: Accessible over the Internet. Example: 8.8.8.8

Private IP Address Ranges

10.0.0.0 – 10.255.255.255

172.16.0.0 – 172.31.255.255

192.168.0.0 – 192.168.255.255

Subnetting: Subnetting divides a network into smaller networks.

Example: Network: 192.x.x.x/24, Subnet Mask: 255.255.255.0

Benefits: Better network management, Reduced broadcast traffic, Improved security

Difference between TCP and UDP

FEATURES	TCP	UDP
connections	Connection oriented: uses a three way hand-shake	Connectionless: no hand-shake
Reliability	Reliable	unreliable
Error checking	yes	Basic
Speed	slower	Faster
Packet ordering	maintained	guaranteed
usage	Web, Email, ssh	Streaming, gaming, VloP

Network attacks

1. **Unauthorized Access Attacks:** These attacks aim to gain access to network systems without permission, either through stolen credentials, brute force, or malicious insiders.
 - Brute-force attacks
 - Credential stuffing
 - Insider threats
2. **Malware-Based Attacks:** Malware disrupts or damages networks through payloads like trojans, ransomware and worms
3. **Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS):** This attack floods network resources with traffic to disrupt availability and functionality.
4. **Man-in-the-Middle (MITM) Attacks:** Here, an attacker secretly intercepts and potentially alters communications between two parties
5. **Spoofing Attacks:** The attacker impersonates a legitimate device or service to deceive systems or users.
 - IP spoofing
 - DNS spoofing
 - DHCP spoofing
6. **Phishing and Social Engineering with Network Implications:** Attackers deceive individuals into revealing credentials that lead to deeper network intrusions.
7. **Advanced Persistent Threats (APTs):** Stealthy, targeted intrusions designed to stay hidden while gathering intelligence or exfiltration data.

Working of Firewall and router

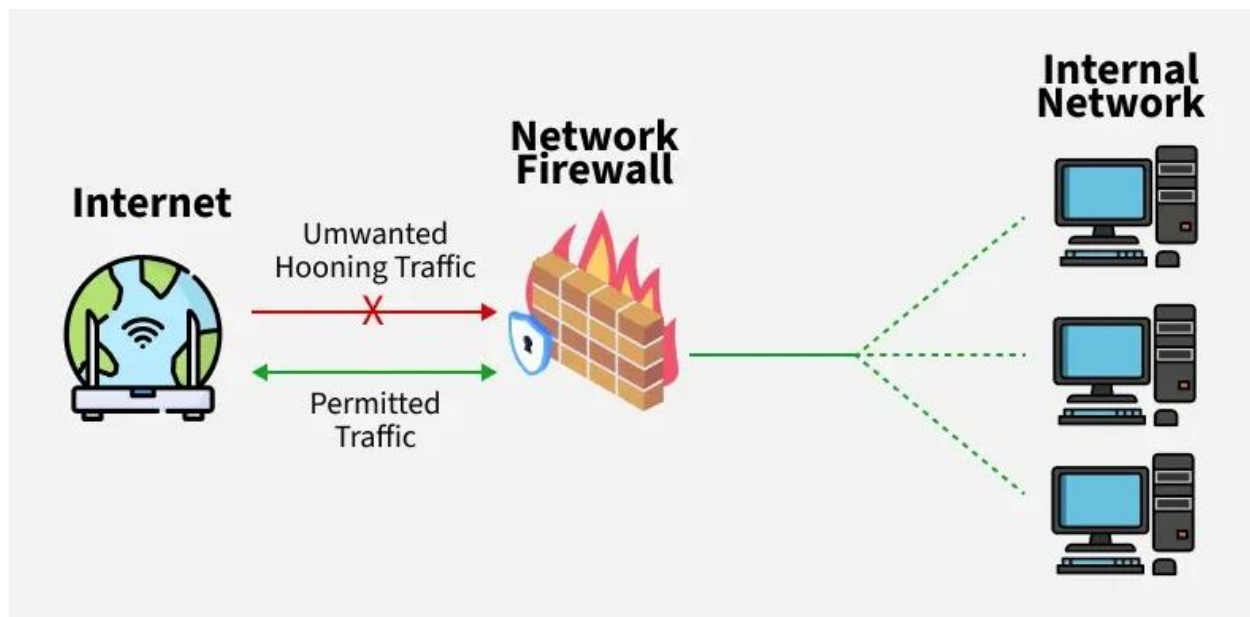
A firewall monitors and controls incoming and outgoing network traffic according to predefined security rules.

Functions:

- Blocks unauthorized access
- Filters malicious traffic
- Monitors network connections
- Enforces security policies

Types:

- Packet Filtering Firewall
- Stateful Firewall
- Next-Generation Firewall (NGFW)



Router working:

A router connects different networks and forwards packets based on IP addresses.

Functions:

- Routes packets between networks
- Connects local networks to the Internet
- Performs Network Address Translation (NAT)
- Reduces unnecessary traffic

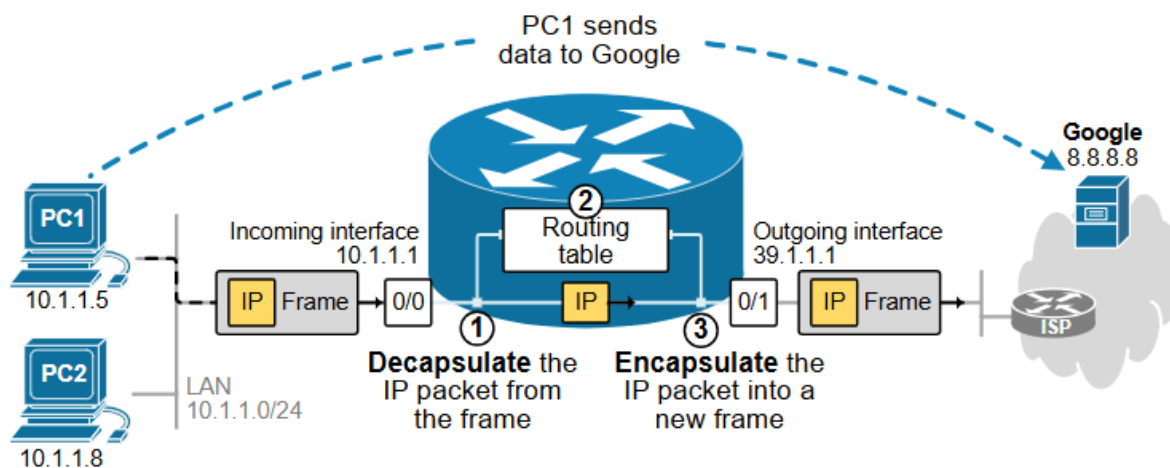


Figure 3. How does a router work?